Labour Supply 2: Labour Supply and Public Policy

EC306- Wages and Employment

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Goals of This Section

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In this section we will:

- Discuss various public policies that affect labour supply
  - Taxes/subsidies, transfers, employment insurance, worker compensation
- Show effect of these policies on work decisions using economic models
- Examine statistical evidence for the model predictions

Income Maintenance Programs

Overview of Income Maintenance Programs

- Represent about 10% of GDP in Canada
- Primary aims:
  - Raise income of specific groups (parents, elderly)
  - Supplement low wages
- Major programs: child benefits, employment insurance, social assistance, pensions

Key Challenge

Balancing income support with work incentives

Government Transfers in Canada

In 2017, 83% of families received some form of government transfer:

Program	% Receiving	Average Amount
Federal Child Benefits	20.2%	$6,314
Employment Insurance	14.0%	$8,352
Social Assistance	11.3%	$8,367
Old Age Security	25.1%	$8,566
CPP/QPP	32.4%	$10,092

Issues in Program Design

Universal vs. Targeted Programs

- Universal: easy to implement but expensive
- Targeted: benefits only those in need but costly to administer

Work Disincentives

- Some programs reduce incentive to work
- Example: welfare benefits can encourage labour force dropout for those with strong preference for leisure

Permanent vs. Transitory Income Loss

- Different situations require different program designs
- Temporary layoff → short-term support
- Long-term disability → longer support + reintegration costs

Think About It

If 83% of Canadian families receive some form of government transfer, why is program design so important? What happens if a program is too generous or too stingy?

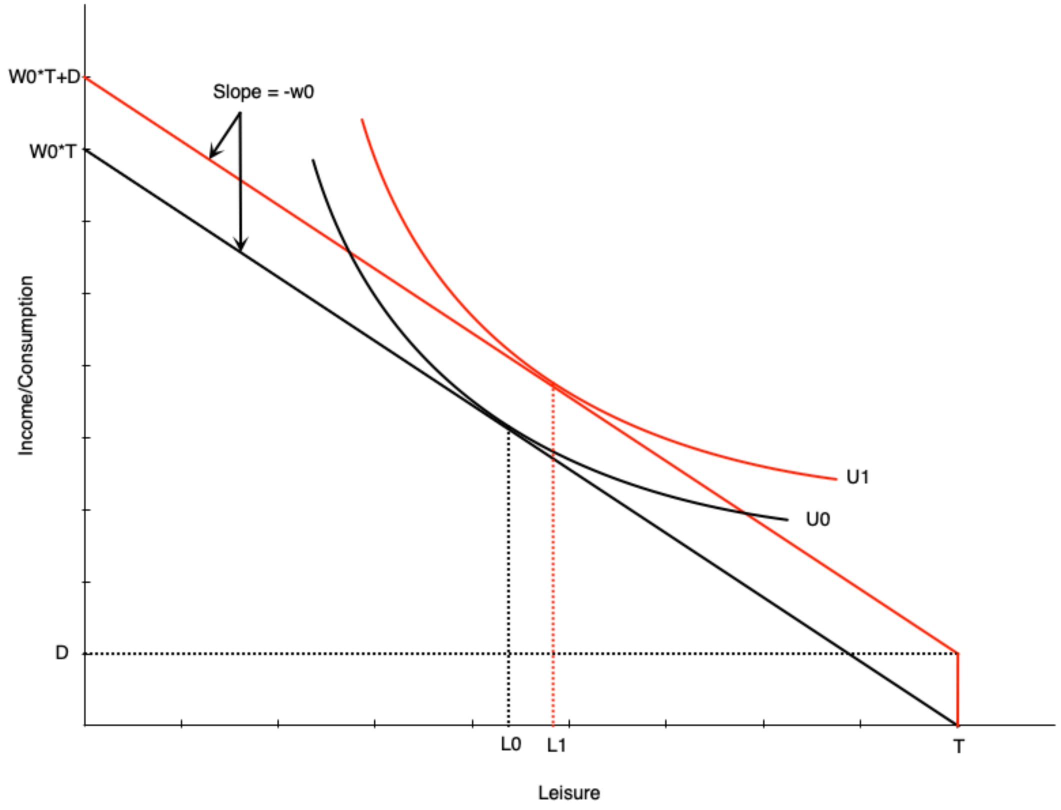
Demogrants, Welfare, Wage Subsidies

Four Standard Income Support Programs

We will examine four policy approaches:

1. Demogrants - Fixed payment to all regardless of income/employment
2. Welfare - Lump-sum payment for non-workers, reduced dollar-for-dollar with earnings
3. Negative Income Tax (NIT) - Welfare payment that decreases at rate < 100% with earnings
4. Wage Subsidies - Addition to the wage rate

Demogrant: Theory



How it works:

- Unconditional lump sum transfer D
- Shifts budget line up parallel
- Wage unchanged → pure income effect

Labor supply impact:

- If leisure is normal → optimal leisure rises
- Reduces hours worked
- May induce non-participation

> ⚠ **Work Incentive Effect**
>
> Unambiguously negative

Demogrant: Canadian Example

Universal Child Care Benefit (UCCB)

- Paid to families with children under 18
  - $160/month per child under 6
  - $60/month per child aged 6-17
- Replaced in 2016 by Canada Child Benefit (CCB)
  - More targeted to low/middle income families

Evidence (Schirle, 2015)

UCCB reduced labour supply of women with children under 6 by ~3.2 percentage points for low-educated married women

Welfare: Overview

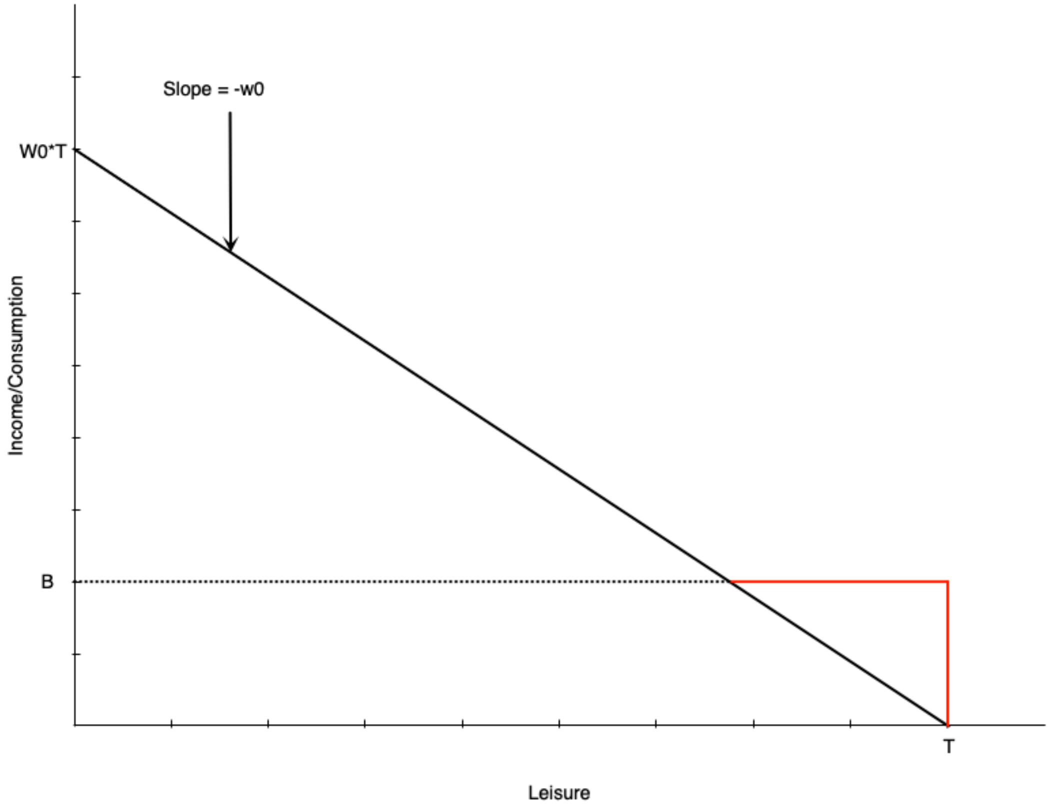
Welfare = Social/Income Assistance in Canada

- Lump-sum payment to non-workers
- Reduced dollar-for-dollar with earned income

Historical Criticism

Creates negative work incentives for some, but must balance against need for income support

Welfare: Budget Constraint



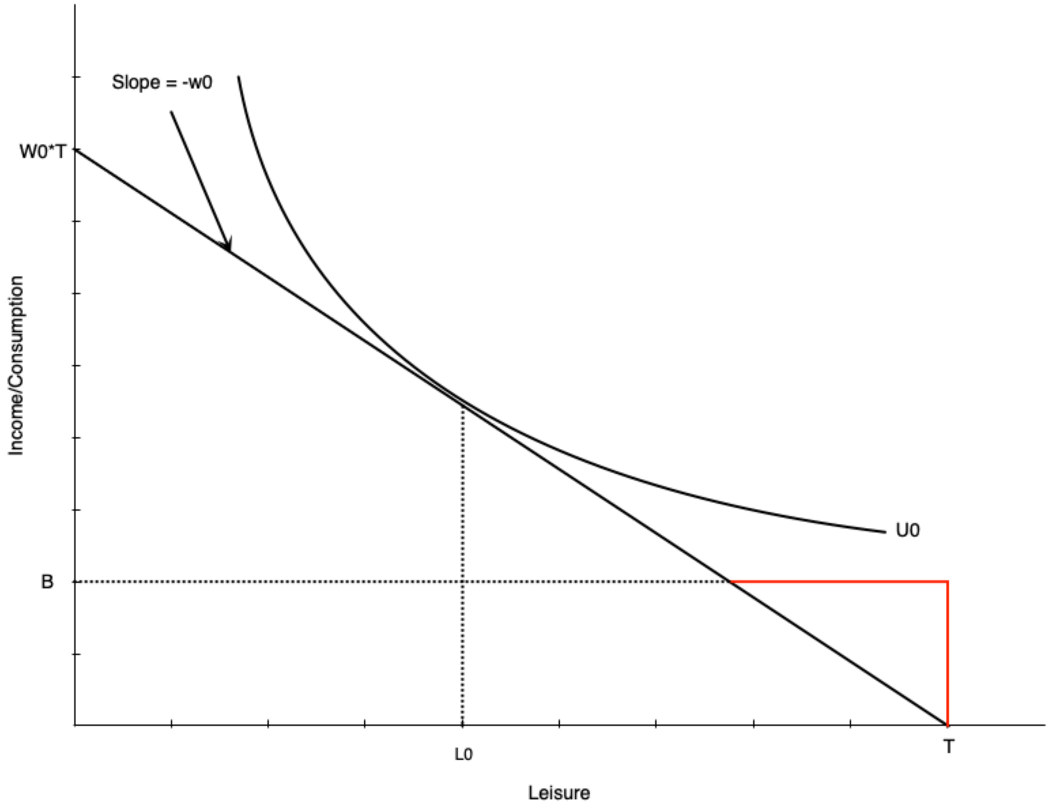
Budget line structure:

- Black line: original budget (leisure at L_0)
- Red then black: with welfare benefit B

Key features:

- Benefit B at zero hours of work
- Reduced \$1 for each \$1 earned
  - Equivalent to 100% tax rate on earnings
- Income constant at B until earned income equals benefit
- After that, earn wage w_0 per hour worked

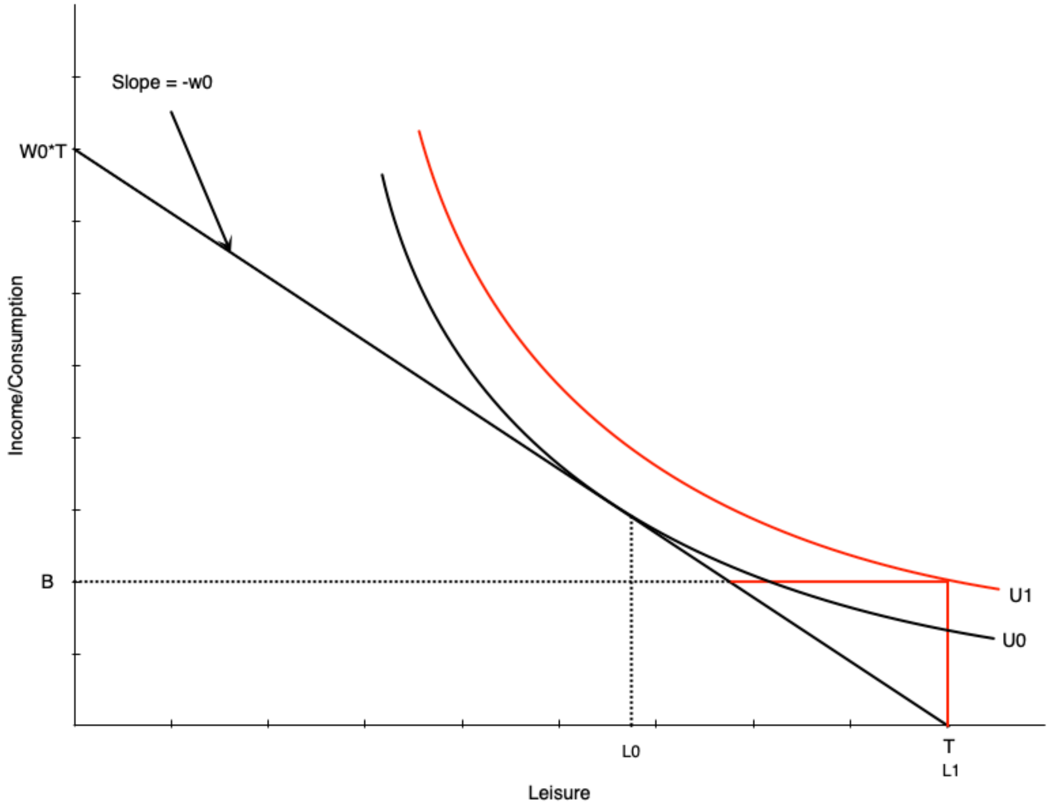
Welfare: No Effect Case



For some workers, welfare has no effect:

- Person working H_0 hours before benefit
- Still maximizes utility at H_0 hours after
- Generally occurs with people already working many hours

Welfare: Work Disincentive Case



Others face strong work disincentive:

- Person chooses zero hours after welfare benefit

This occurs when:

- Welfare benefit is generous
- Individual places high value on leisure

> ⚠ **Warning**
>
> This is the problematic effect of welfare programs

Discussion

Why might someone who was working 20 hours per week before welfare was introduced choose to stop working entirely, while someone working 40 hours per week continues to work the same amount?

Combating Work Disincentives

Policy options:

1. Reduce benefit level
   - But may not provide adequate support
2. Increase wages
   - Through minimum wages, subsidies, training, unions
   - Can be expensive and may reduce labour demand
3. Change preferences
   - Alter attitudes toward work/leisure
   - Difficult to achieve
4. Negative income tax
   - Reduce tax-back rate on earnings
   - Discussed next…

Welfare: Evidence from Quebec

Natural experiment: Welfare benefit structure changed in 1989

Before 1989:

- Age-based benefits for singles/couples with no kids
- Under 30: $185/month
- 30 and over: $507/month

Results at age 30:

- Much longer time on welfare
- Lower employment rates

Note

Turning 30 created a large discontinuous jump in benefits

Negative Income Tax: Overview

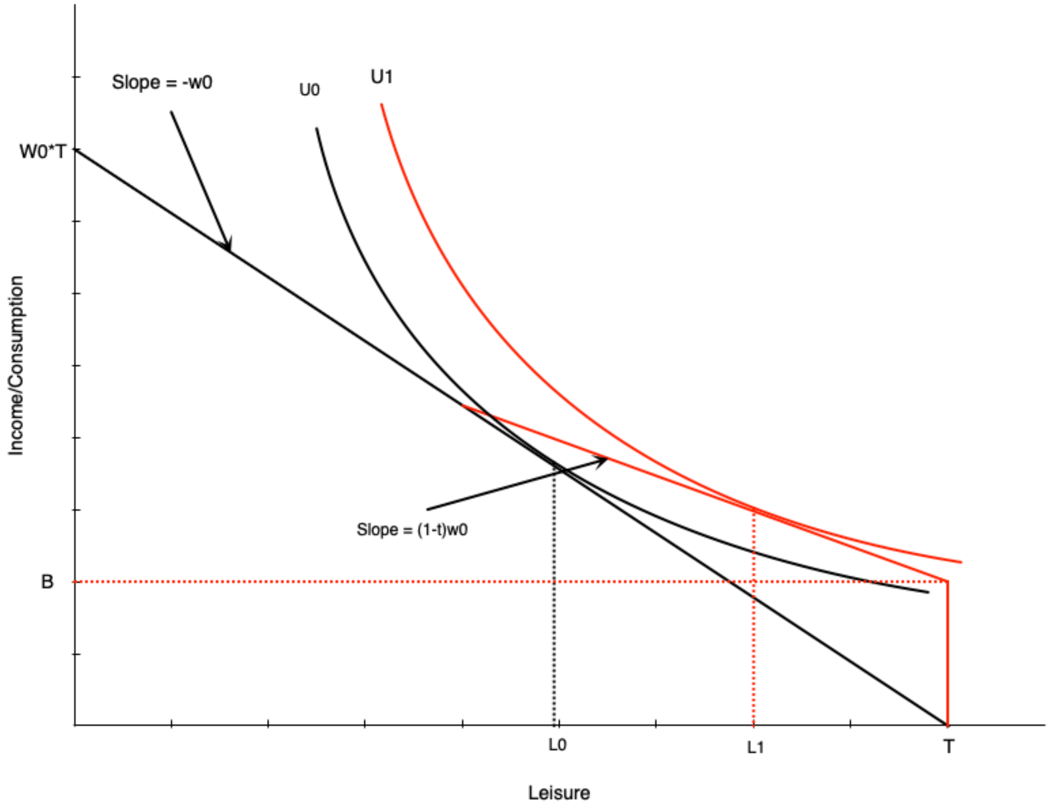
What is NIT?

- Welfare program where benefit reduced at rate $< 100\%$ of earnings
- Example: $10,000 benefit reduced by 50¢ per dollar earned

> **Advantages**
>
> Provides income support with fewer work disincentives. Still maintains incentive to work since benefit reduces slowly.

Also known as: Guaranteed Annual Income (GAI) or Basic Income

Negative Income Tax: Budget Constraint



NIT has two components:

- Benefit B at zero hours of work
- Tax rate t on earned income (until earned income equals benefit)

Effect on labour supply:

- Pivots budget line up vs. pure welfare
- Work hours reduced from H_0 to H_1
- Person still works some hours

> **Important**
>
> Work disincentive less than pure welfare

Negative Income Tax: Income and Substitution Effects

Income effects:

- Benefit increases non-labour income → increases leisure ↑
- Tax has negative income effect → reduces leisure ↓
- Net: benefit effect dominates → leisure ↑

Substitution effect:

- Tax rate reduces effective wage
- Makes leisure relatively cheaper → reduces work hours ↓

Overall Effect

Negative effect on work hours, but smaller than pure welfare

Negative Income Tax: Math

The budget constraint with NIT:

$$C = \begin{cases} wh + B - twh, & \text{if } B - twh \geq 0 \\ wh, & \text{otherwise} \end{cases}$$

Interpretation:

- Consumption = earned income (wh) + net benefit ($B - twh$) if net benefit > 0
- Otherwise just earned income (wh)

Break-even point: benefit reaches zero when $wh = \frac{B}{t}$

MINCOME: Manitoba's NIT Experiment

Program details (1970s):

- Randomized controlled trial
  - Treatment group received NIT
  - Control group did not
- Conducted in Dauphin (saturated - everyone got NIT) and Winnipeg
- 7 treatment groups with different benefit levels and tax rates
- Ended in 1979 due to political changes

MINCOME Treatment Groups

Guarantee at enrolment	Tax Rate (on total Income) 35%	50%	75%
$3800	Plan 1	Plan 3	*
$4800	Plan 2	Plan 4	Plan 7
$5480	X	Plan 5	Plan 8

Plan 9

* Plan 6 was collapsed into Plan 7 because participants viewed this option as too punitive and left the experiment.
X This option was deemed to expensive for the budget.
Plan 9 was the control group

Source: Wikipedia

MINCOME: Results

Labour supply effects:

- Small decline in hours worked
  - Mainly mothers of newborns
  - Women in two-parent families
  - No significant decline for men

Health outcomes:

- Fewer hospitalizations and doctor visits
- Especially for mental health and accidents

Education:

- Increase in adolescents finishing high school

Wage Subsidies: Theory

What is a wage subsidy?

- Payment to workers that increases their wage rate
- Example: \$5/hour subsidy increases \$15/hour wage to \$20/hour

Effect on budget constraint:

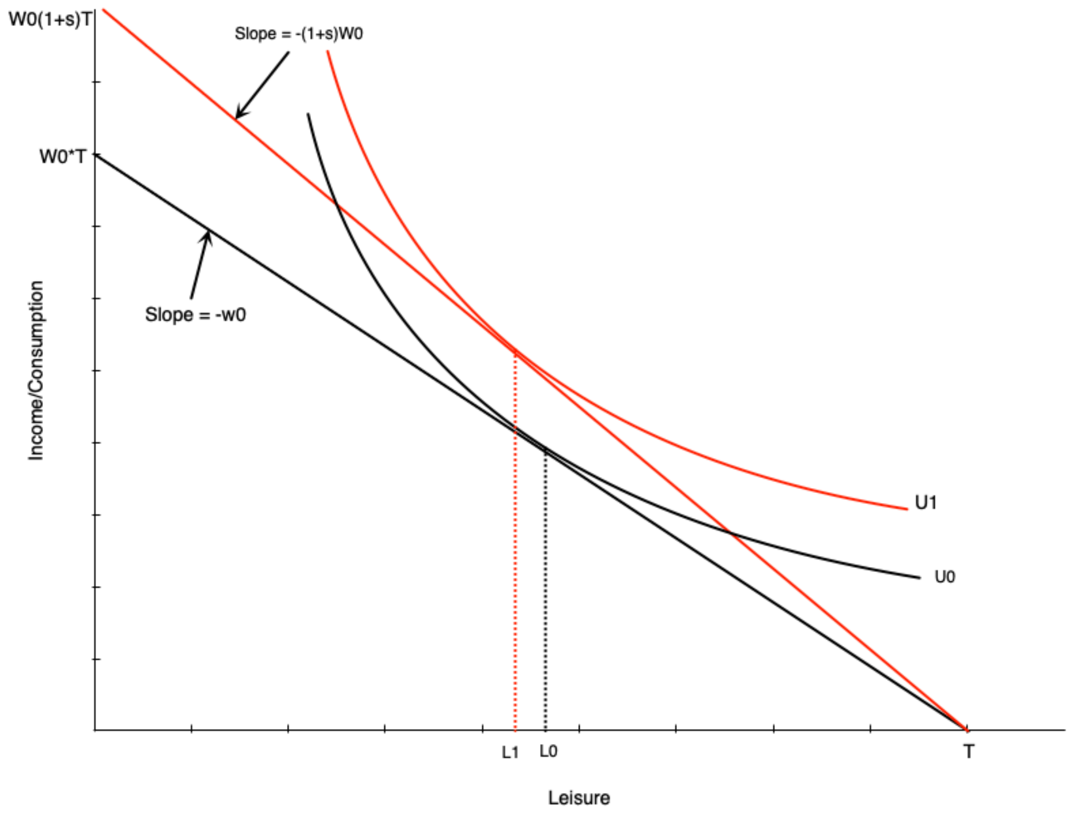
- Budget line pivots upward

Labour supply effects:

- Income effect → increases leisure ↑
- Substitution effect → decreases leisure ↓

> ⚠ **Warning**
>
> Overall effect on work hours is ambiguous

Wage Subsidies: Diagram



Budget line pivots up with subsidy

- Graph shows person working more
- But effect is theoretically ambiguous
- Income and substitution effects work in opposite directions

Wage Subsidies: Policy Implications

Advantages

- Less negative work incentives than welfare or NIT
- No reduction in effective wage rate

Disadvantages

- Do not provide income support to non-workers
- May not help those in most need

In Canada:

- Pure wage subsidies mostly go to employers
- Unclear if any benefits passed on to workers

Think About It

A negative income tax reduces the benefit as you earn more, while a wage subsidy increases your pay for each hour worked. Which program would you expect to have a larger positive effect on labour force participation among non-workers, and why?

Tax Credits: Overview

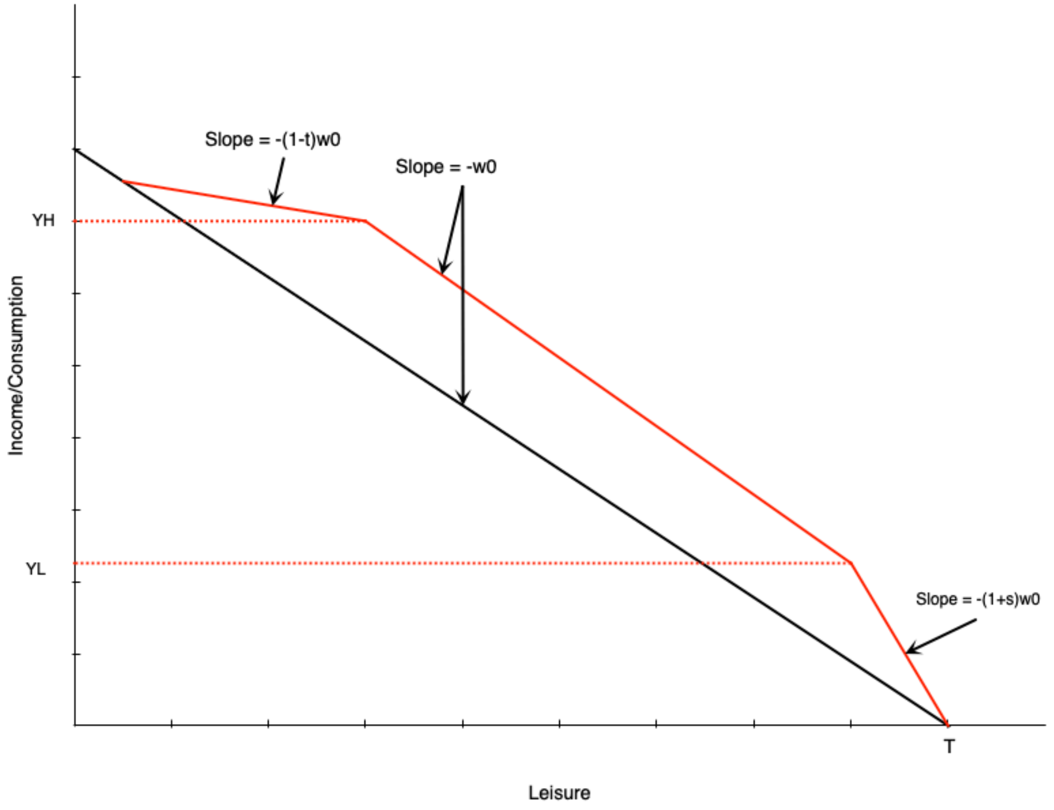
Two types of tax credits:

1. Non-refundable: Can reduce tax owed to zero (but not below)
2. Refundable: Can reduce tax owed below zero (results in payment)

Income Maintenance Programs via Refundable Tax Credits

- Canada Worker Benefit (CWB) - Canada
- Earned Income Tax Credit (EITC) - US

Tax Credits: Budget Constraint



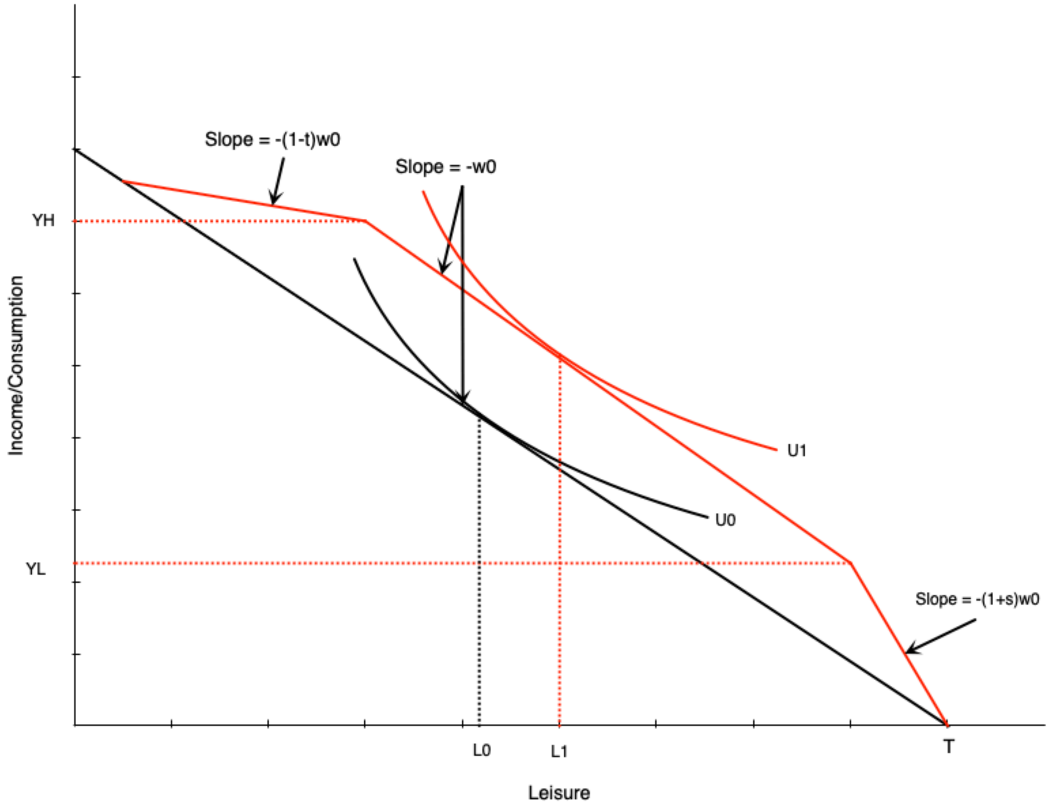
Three segments:

1. Phase-in: Subsidy proportional to earned income
2. Plateau: Maximum benefit level
3. Phase-out: Benefit reduced with earned income

Combines elements of:

- Wage subsidy (phase-in)
- Demogrant (plateau)
- NIT (phase-out)

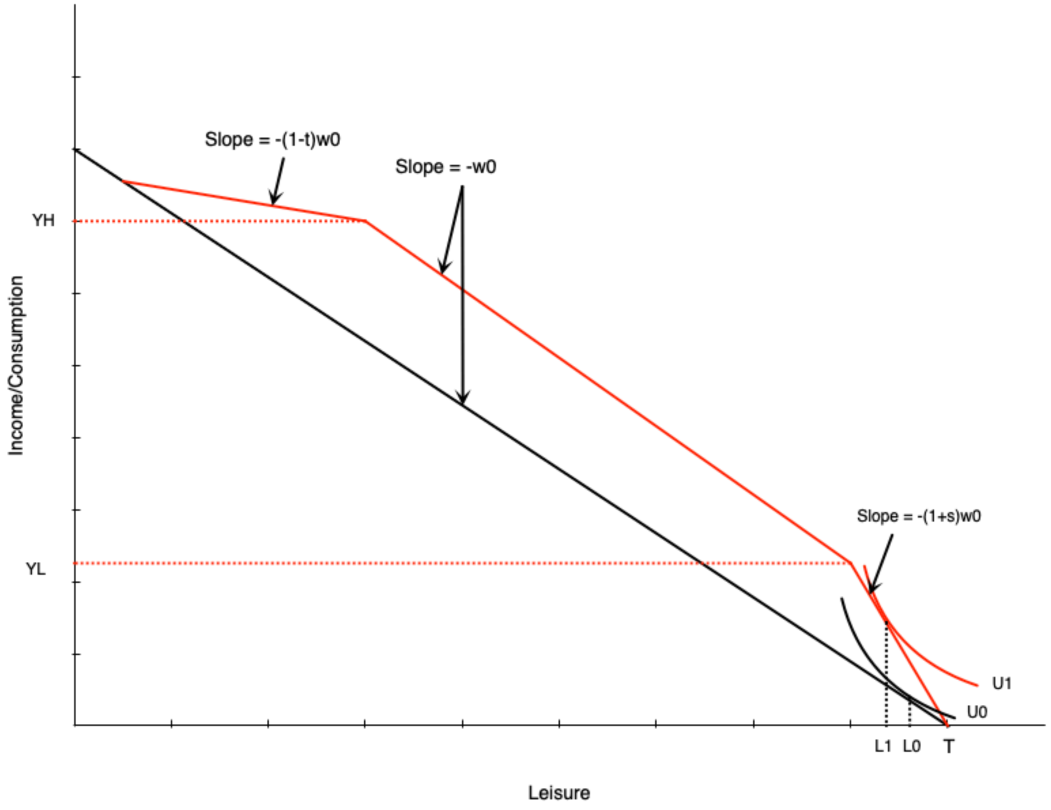
Tax Credits: Effect in Plateau Range



Person initially in middle range:

- Tax credit gives maximum benefit
- Acts like pure income effect
- If leisure is normal good → reduce work hours

Tax Credits: Effect in Phase-In Range



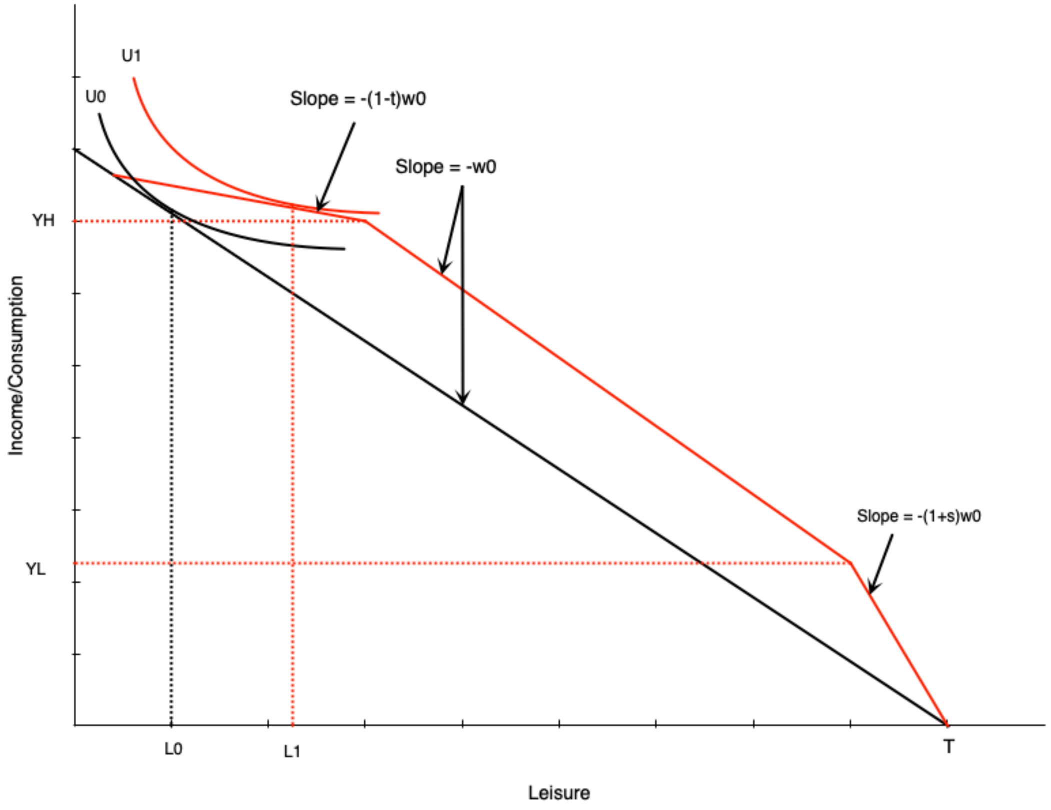
Person in phase-in range:

- Sees increase in effective wage
- Acts like pure wage subsidy
- Income and substitution effects in opposite directions

⚠ **Warning**

Effect on hours is ambiguous

Tax Credits: Effect in Phase-Out Range



Person in phase-out range:

- Similar to negative income tax
- Income effect → reduce hours ↓
- Substitution effect → reduce hours ↓

> ❗ **Important**
>
> Effect on work is unambiguously negative

Unemployment Insurance

Employment Insurance: Overview

Purpose:

- Safety net for workers who lose their job
- Collect income while unemployed

Funding:

- Payroll taxes on employers and employees

Two main components:

1. Regular benefits - for those unemployed through no fault of their own
2. Special benefits - sickness, maternity, parental leave, caregiving

Employment Insurance: Benefits

Regular Benefits Structure

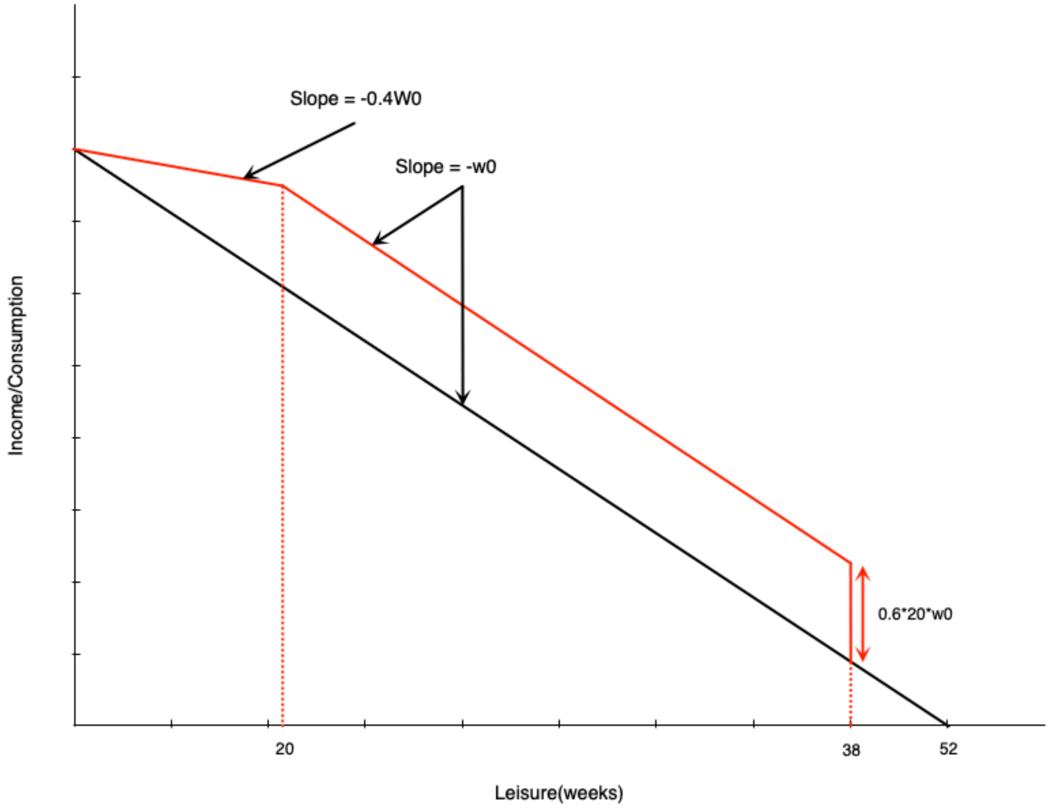
- 55% of average insurable weekly earnings
- Up to a maximum amount
- Duration: 14 to 45 weeks
  - Depends on unemployment rate and hours worked in last year

EI: Simplified Model Parameters

To analyze with income-leisure diagram:

- Must work 14 weeks in the year to qualify
- Benefit is 60% of weekly wage (we use 60% for simplicity)
- Maximum of 20 weeks of benefits
- Can work up to 32 weeks and get full benefits
- Each week worked beyond 32 → lose benefits for that week
- 60% benefit = 60% tax on earnings beyond 32 weeks
- Cannot work and collect EI simultaneously

EI: Budget Constraint



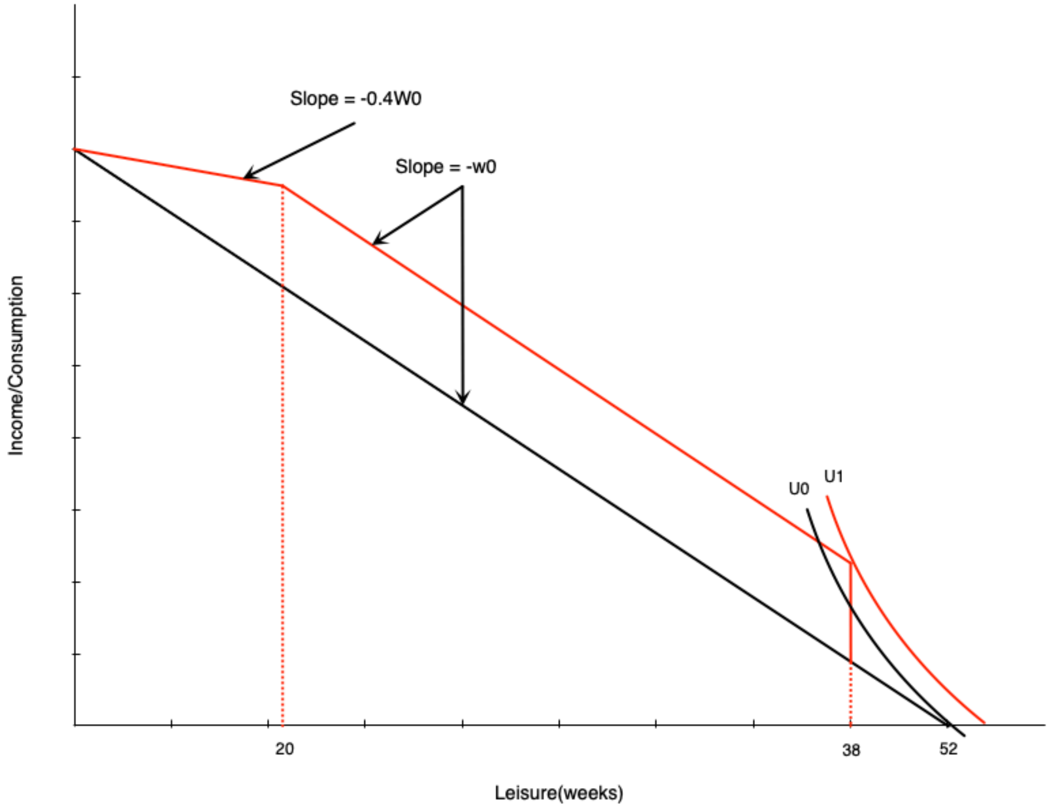
Three segments:

1. First 14 weeks: Paid at usual wage w_0
2. Weeks 14-32 of work: Get 20 weeks of benefits
   - Benefit = $0.6 \times 20 \times w_0$
3. More than 32 weeks of work: Benefits reduced week-for-week
   - Benefits paid at 60% of wage
   - Work paid at 100% of wage
   - Keep 40% of wage in this range

Discussion

The EI budget constraint has a segment where the effective wage is only 40% of the market wage. Why might a seasonal worker in the Maritimes choose to work exactly 14 weeks per year under this system?

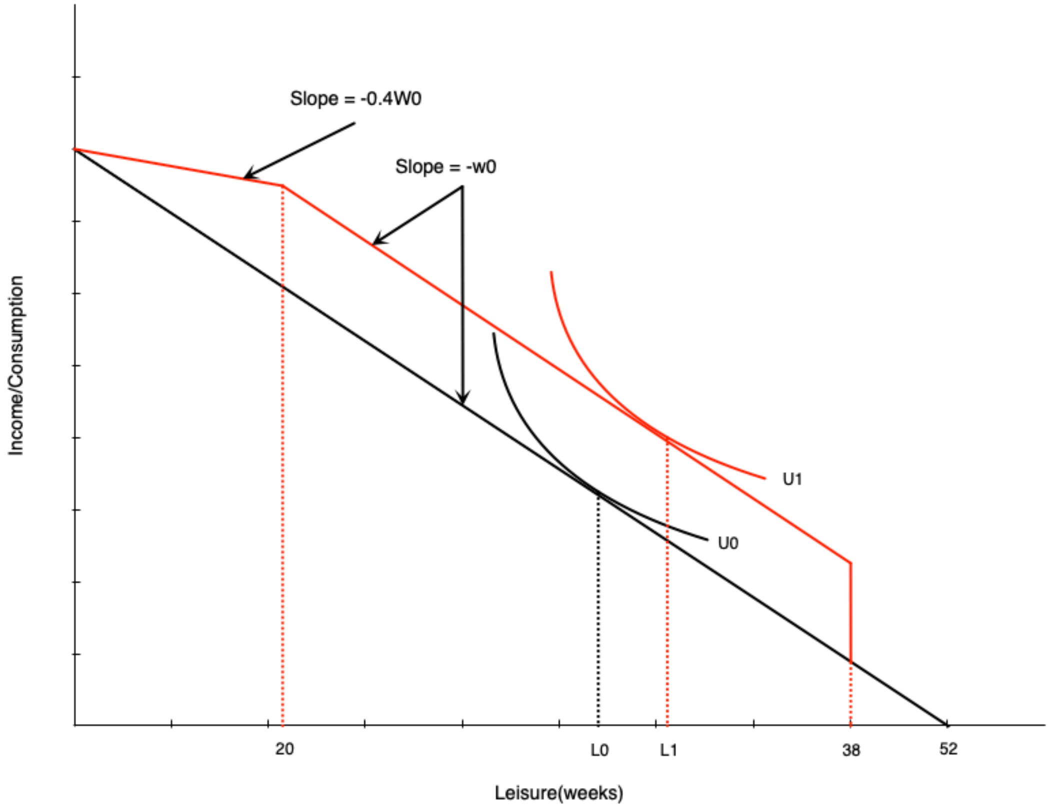
EI: Effect on Non-Participants



Effect on Non-Workers

A non-participant might start working due to large increase in non-labour income after 14 weeks of work. Induces labour force entry.

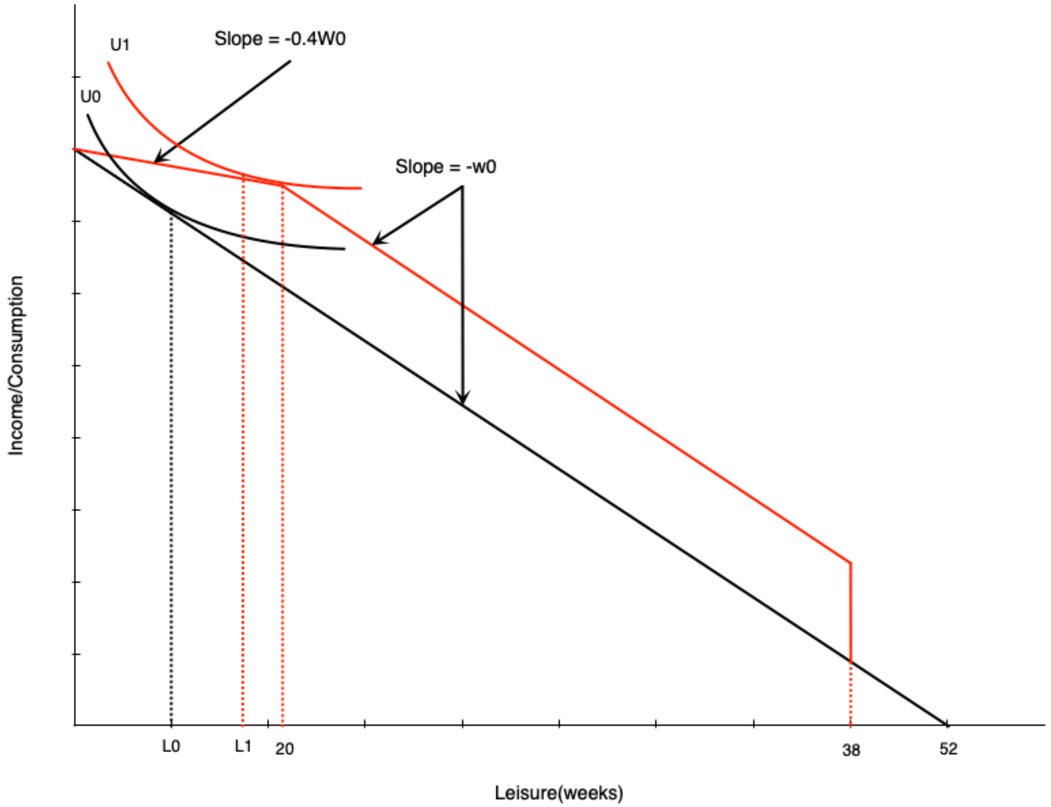
EI: Effect in Middle Range



Person getting full benefits:

- Acts like pure income effect
- Will reduce work weeks

EI: Effect on High-Hour Workers



In phase-out range:

- Acts like negative income tax
- Income effect → reduce hours ↓
- Substitution effect → reduce hours ↓

Important

Effect unambiguously negative

Evidence: Maine vs. New Brunswick

Natural experiment:

- Initially had very similar EI schemes
- Later, New Brunswick increased benefit generosity relative to Maine
- Difference-in-differences approach:
  - Compare changes in NB before/after to Maine over same period
  - Maine acts as control group
  - Controls for other time-varying factors

Evidence: Maine vs. New Brunswick Results

New Brunswick experienced:

- Higher labour force participation for marginally-attached workers
- Lower hours for those with stronger attachments

> **Interpretation**
>
> - Corresponds exactly to model predictions
> - Highlights work disincentive effects of income maintenance programs

Disability and Workers Compensation

Introduction

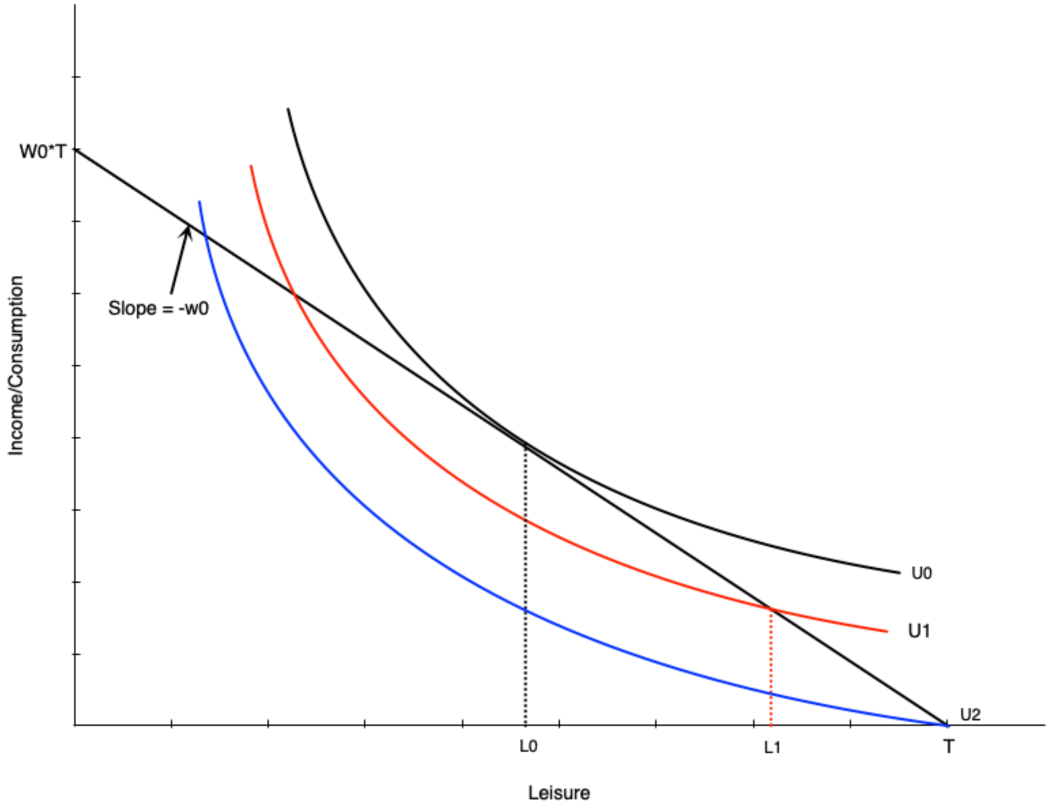
What are these programs?

- Provide income support to injured/ill workers
- Come from public and private sources
- Premiums paid by employers and employees

Work Incentive Considerations

- Severe injury/disability: work incentives not a concern (unable to work)
- Partial disability: people may still be able to work

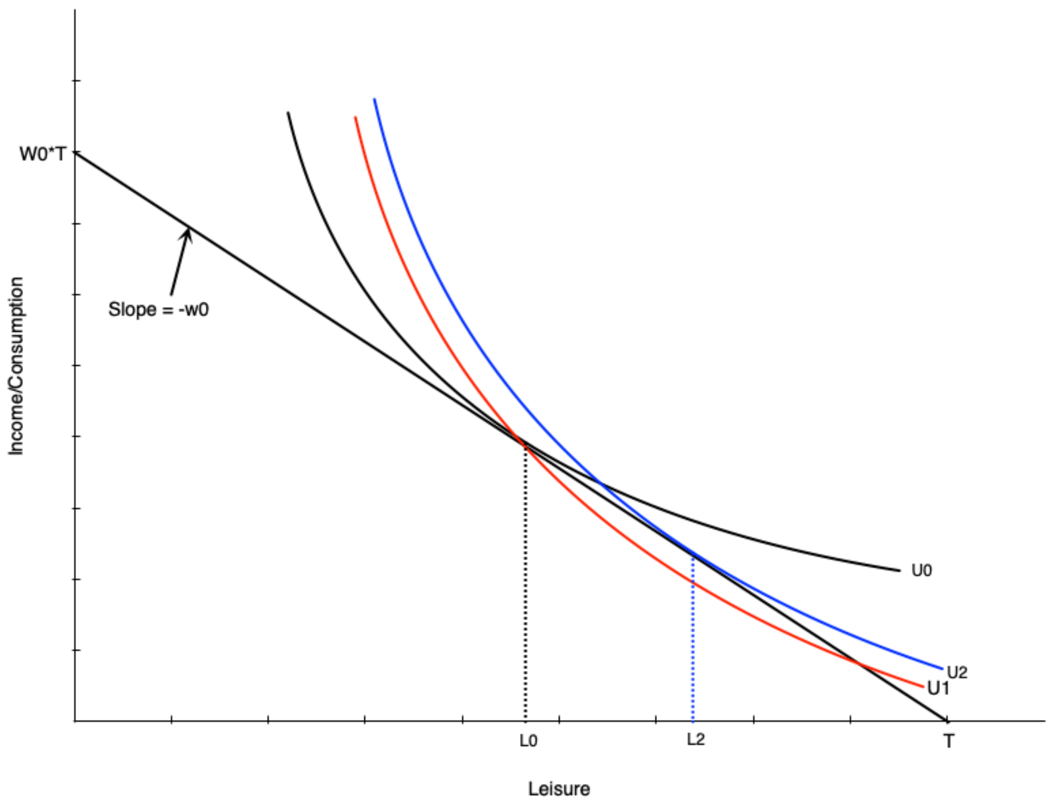
Effect of Disability on Labour Supply



Three scenarios:

1. Initially: Choose L_0 leisure, work H_0 hours
2. Injured but can work: Choose L_1 leisure, work H_1 hours
   - Lower (non-optimal) indifference curve
3. Completely disabled: Choose T leisure, zero work
   - Even lower indifference curve

Disability and Preferences



Disability may change preferences:

- Work becomes more difficult
- Preferences tilt toward leisure
- Indifference curve becomes steeper
- MRS increases at each point
- More willing to give up consumption for leisure

Result:

- Even if unconstrained, choose more leisure
- Person moves from L_0 to L_2

Compensation Schemes

Typical disability insurance structure:

- Pays a fraction of previous earnings

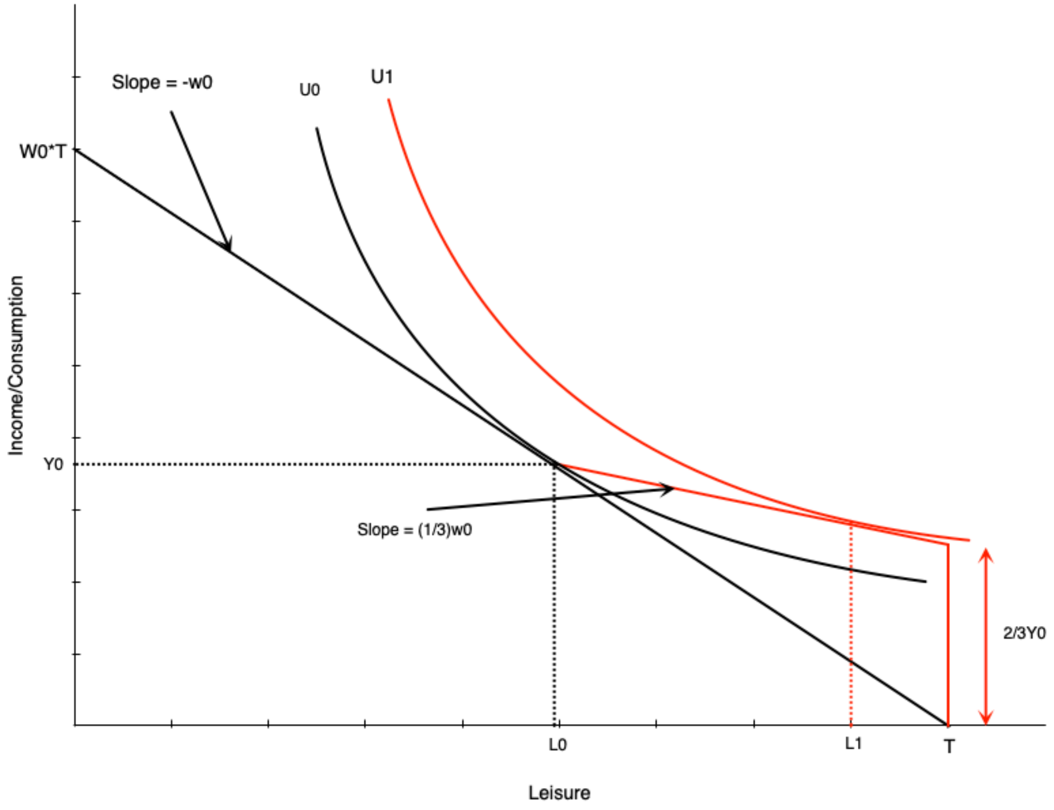
Example: Ontario WSIB

- 85% of net earnings (up to maximum)
- Lasts until return to work or age 65

Two schemes we'll examine:

1. Two-thirds income replacement
2. Full income replacement

Two-Thirds Compensation



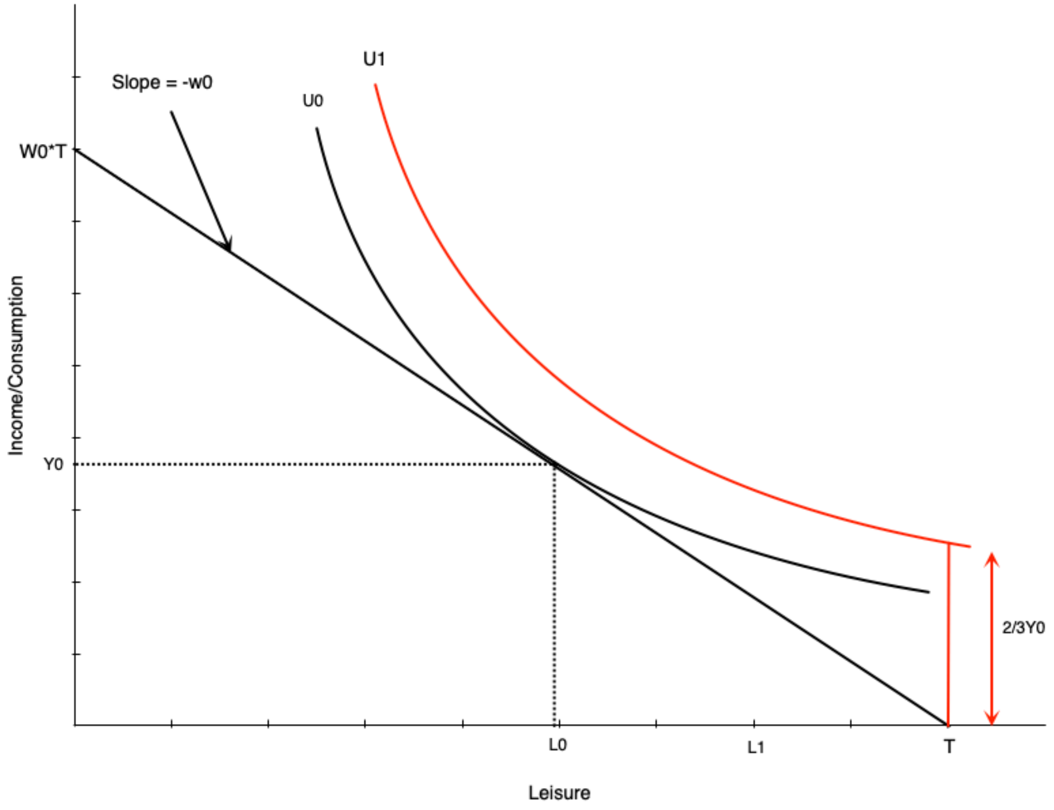
Looks like negative income tax:

- At zero hours: get 2/3 of previous income
- For each hour worked:
  - Gain full wage
  - Lose 2/3 of wage in benefits
  - Keep 1/3 of wage

For workers with flexible hours:

- Effects match NIT
- Negative effect on hours worked

Restricted Hours: Generous Compensation



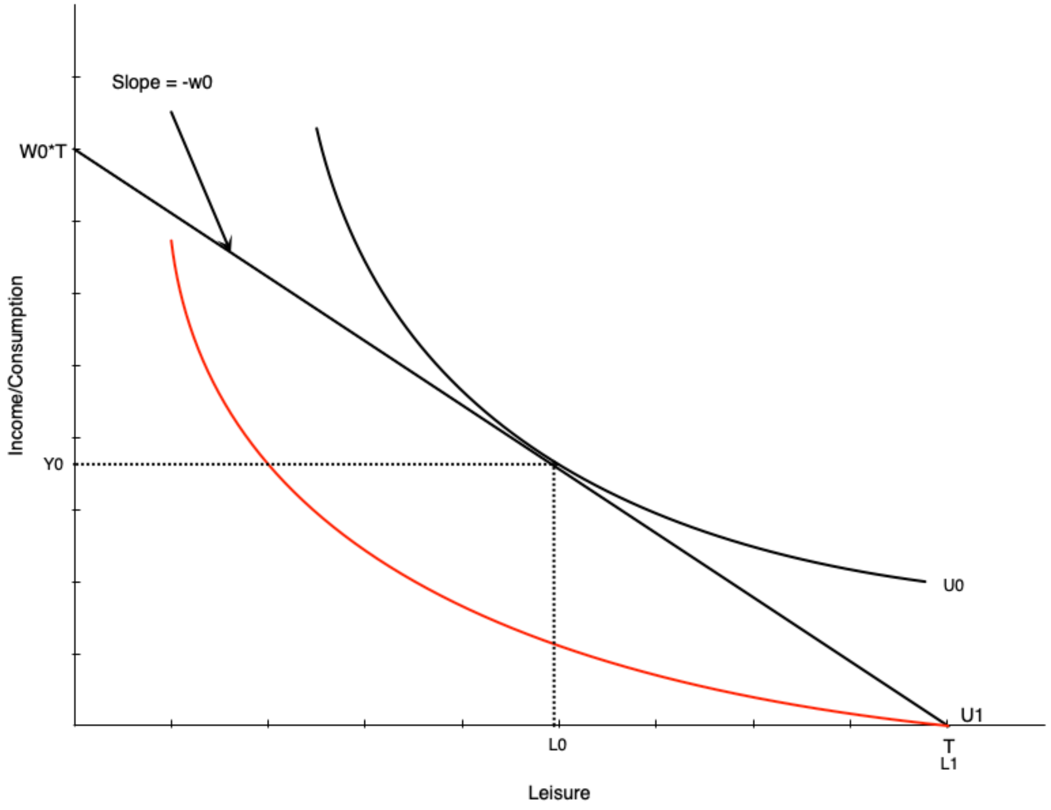
What if worker can only work H_0 hours or zero?

- If scheme generous enough → choose zero hours
- Utility at zero hours > utility at H_0 hours

Implication

Strong work disincentives of generous scheme

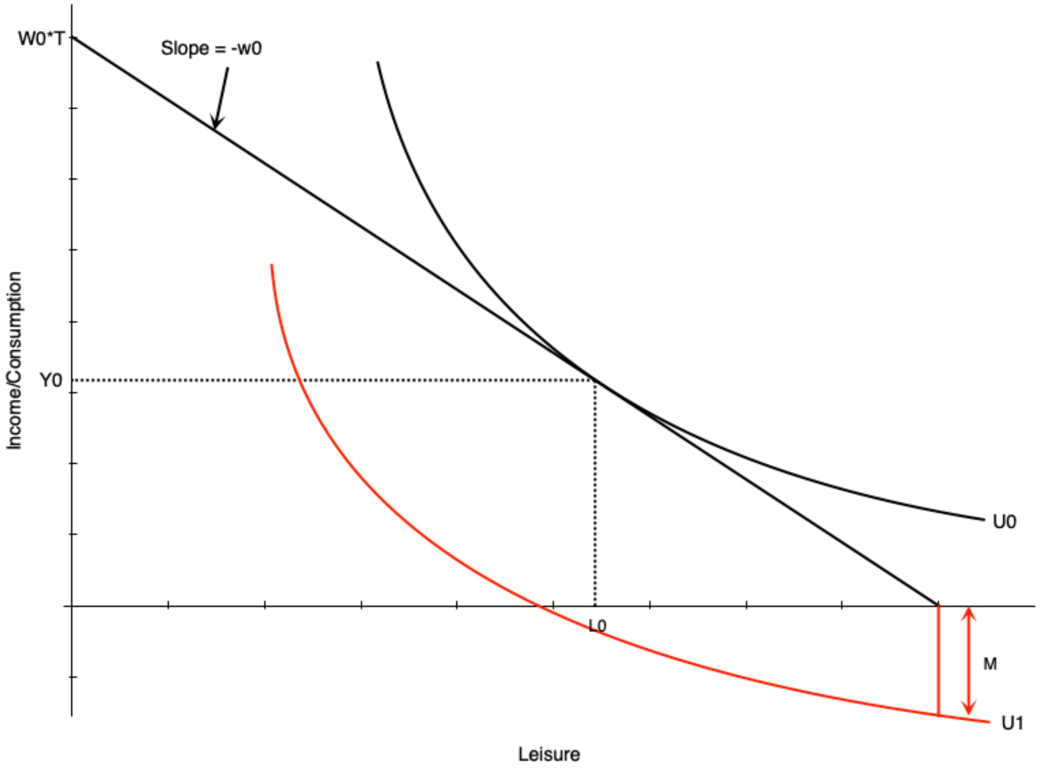
No Compensation Case



Not Providing Compensation is Problematic

- Person who can work chooses H_0 hours
- Person who cannot work gets lower utility
- End up substantially worse off

Medical Expenses Case

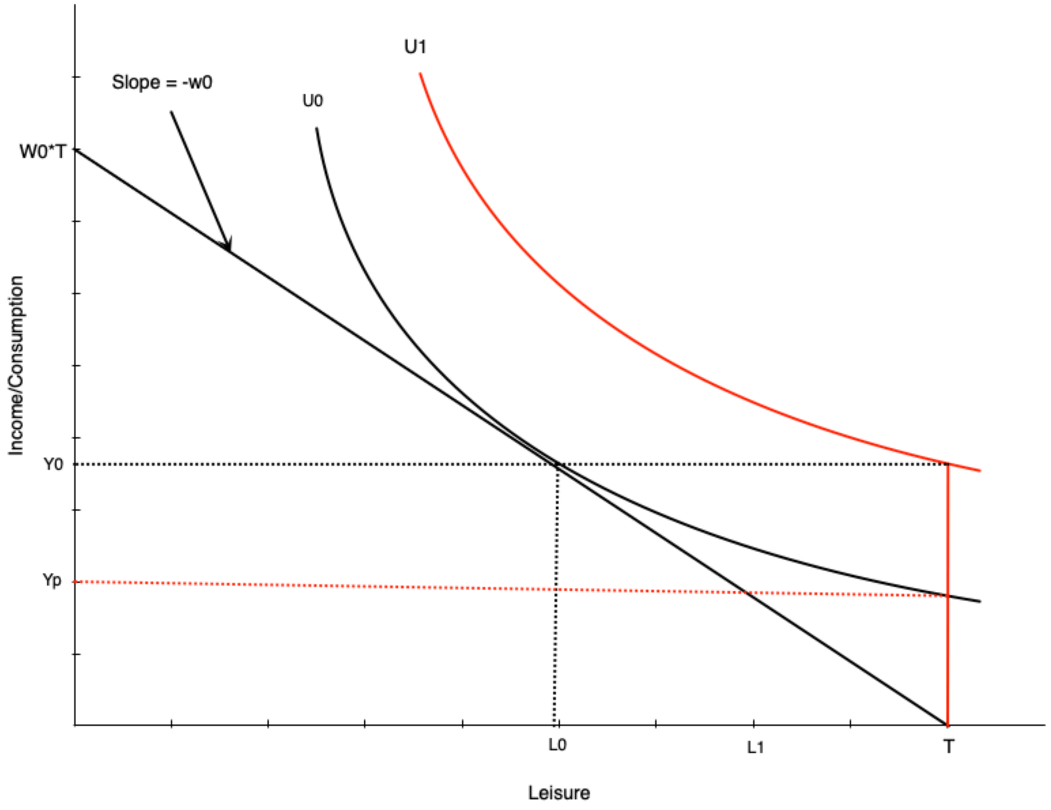


Medical expenses worsen situation:

- Act like reduction in non-labour income
- If large enough, non-labour income could go negative
- Occurs with no/very low compensation

This is an extreme scenario

Optimal Compensation Level



How much to compensate non-workers?

Two options:

1. Full income replacement
    - Higher indifference curve (U_1) than working
2. Same satisfaction as working
    - Same indifference curve as working
    - Lower compensation (Y_p)

ⓘ Note

The optimal level is a policy question

Child Care Subsidies

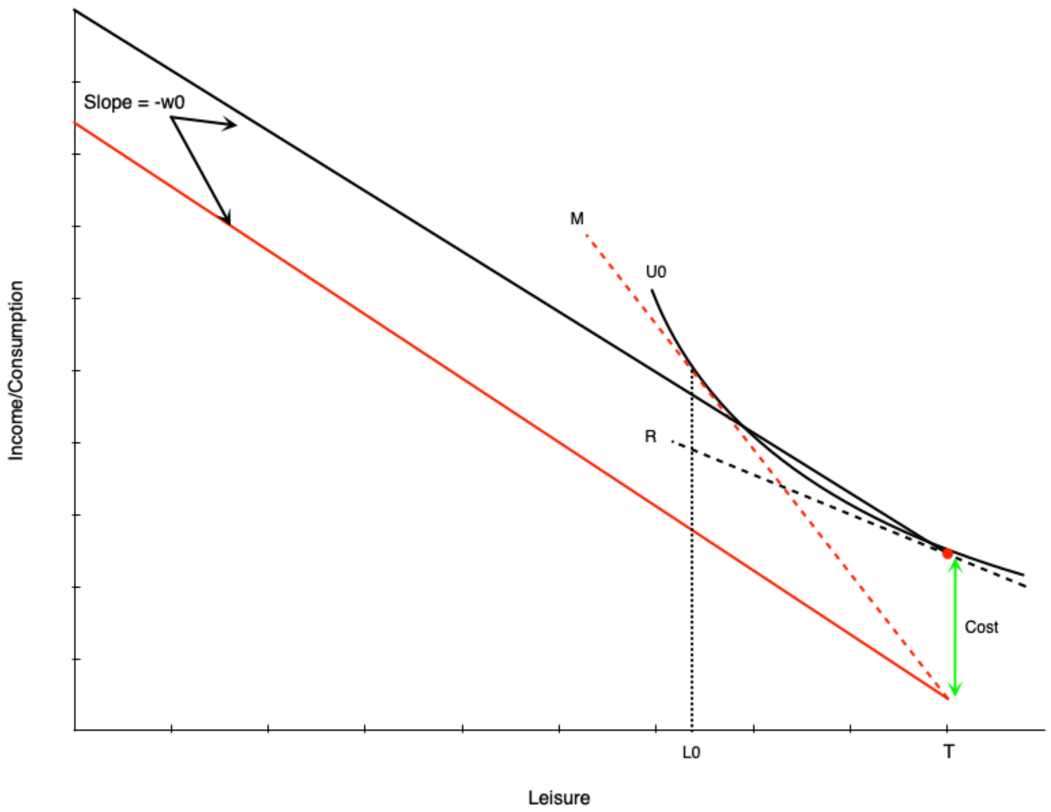
Introduction

Canada’s child care expansion:

- Goal: reduce average cost to $10/day by 2026
- More than half of provinces already at $10/day average
- Subsidy varies substantially by province
- Requires provincial-federal agreements

> **Quebec Case Study**
>
> Had $10/day child care since 1997. Provides useful evidence on effects.

Child Care Costs: Non-Working Case



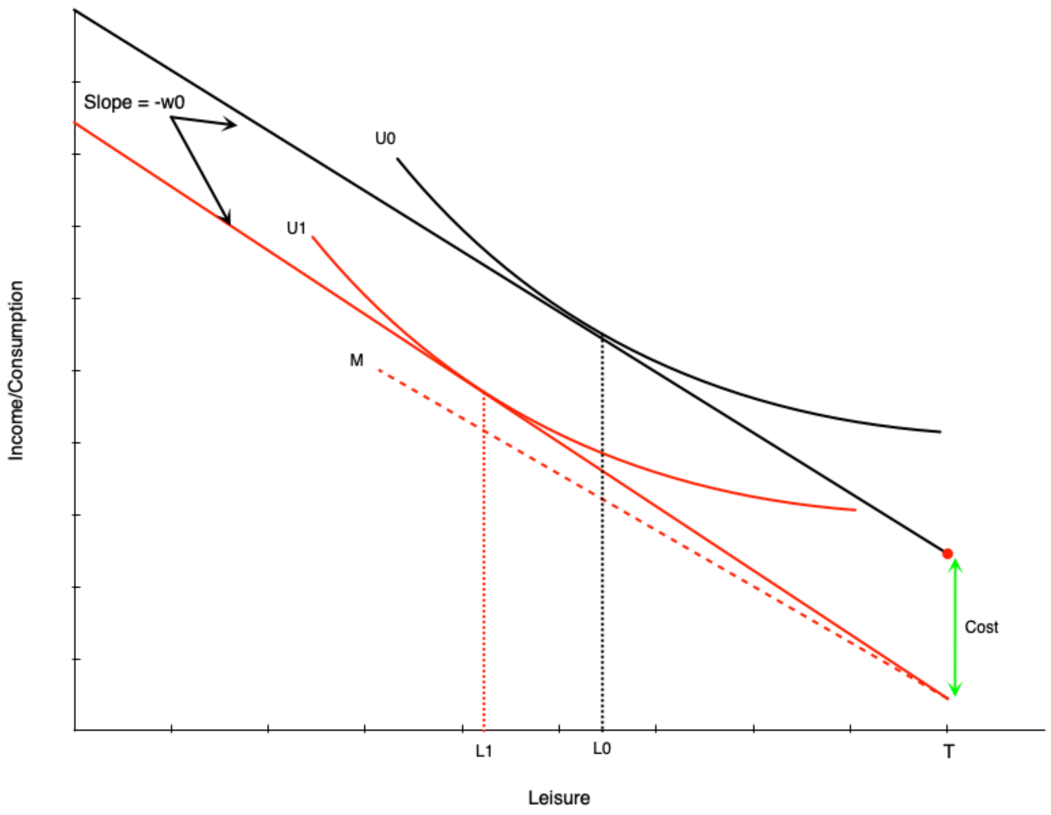
Budget line becomes discontinuous:

- At zero hours: red dot
- For any work: drops by child care costs
- Then straight line with slope = wage

Effect:

- Child care costs increase reservation wage
- Need higher wage to induce work
- This person chooses not to work

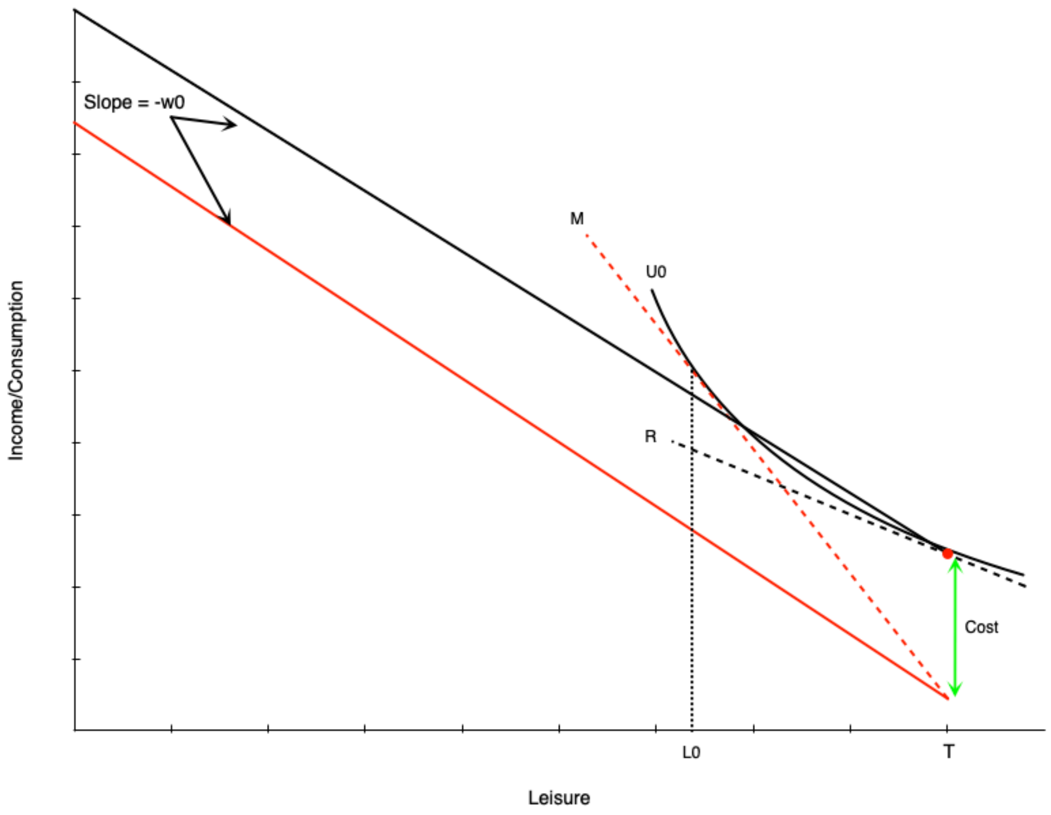
Child Care Costs: Working Case



Person with reservation wage below wage + costs:

- Works positive hours
- Child care costs = negative pure income effect
- Choose less leisure, more work
- Must work more to offset income loss

Child Care Costs: Labour Supply Gap



Creates Gap in Labour Supply Schedule

- Person who works will work minimum of $H_0 = T - L_0$ hours
- Not worth working less due to large child care costs
- Labour supply schedule jumps from zero to H_0 hours

Child Care Subsidies: Effects

A full subsidy (costs → zero) would:

1. Move budget line up in parallel
2. Reduce reservation wage
    - Lower barrier → encourages participation
    - Especially encourages part-time work
3. Create positive income effect
    - Increases consumption of leisure
    - Reduces work hours

Evidence: Quebec's $5/Day Program

Program details:

- Implemented in 2000
- Currently sliding scale: $8.25 to $21.45

Research design:

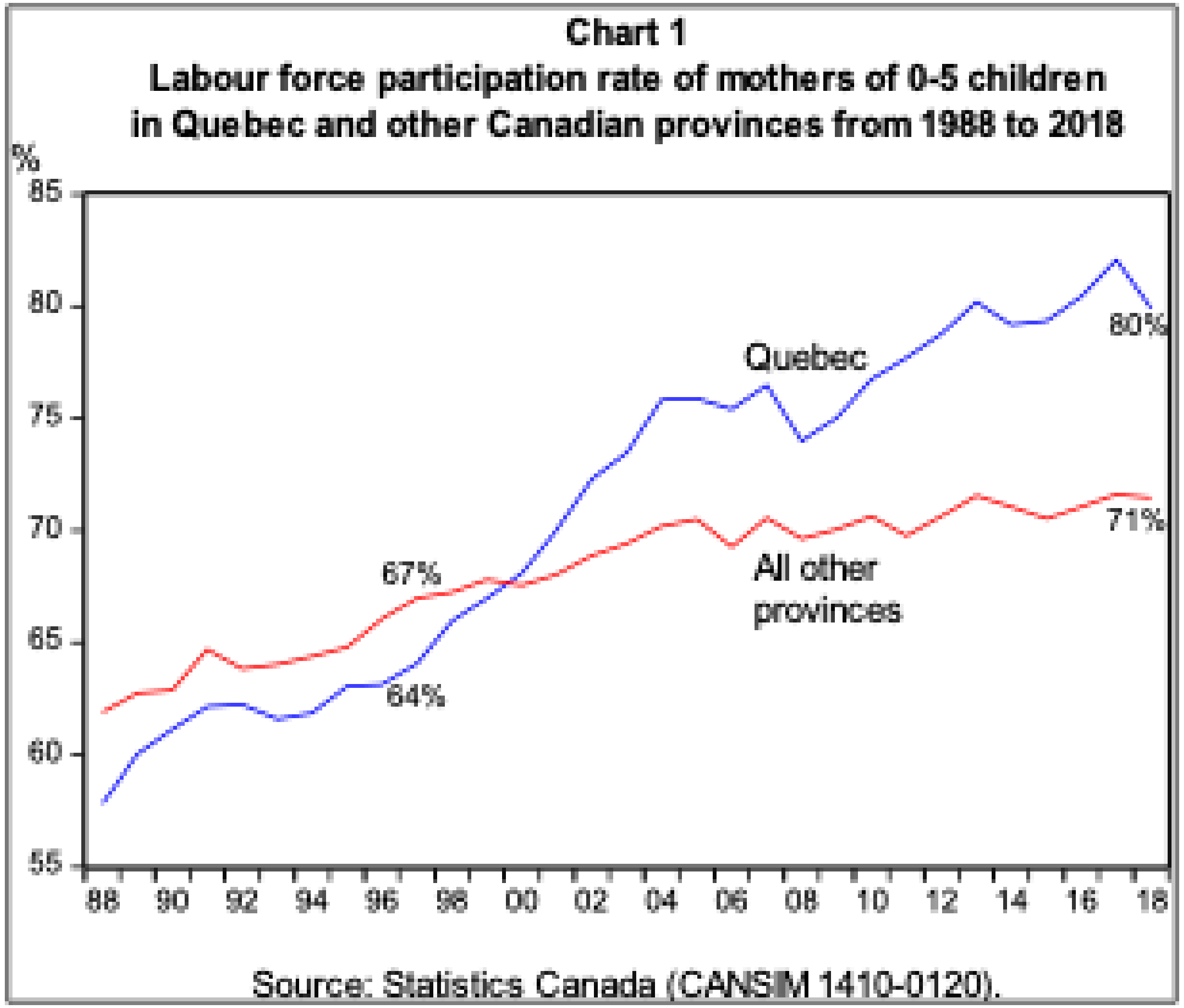
- Difference-in-differences approach
- Compared Quebec mothers to mothers in other provinces
- Other provinces = control group

> **Key Finding**
>
> Labour force participation of mothers with young children ↑ 7.7 percentage points

Evidence: Quebec's $5/Day Program

> ⚠ **Warning**
>
> Unfortunately also found negative effects on child mental health, non-cognitive behaviour, crime later in life

Evidence: Visual Summary



Source: Fortin (2019)

Think About It

Quebec's $5/day child care program increased mothers' labour force participation by 7.7 percentage points but was associated with negative child outcomes. What does this trade-off suggest about how we should evaluate the success of a child care subsidy policy?

Summary

Key Takeaways

Income maintenance programs:

- Provide crucial income support to various groups
- Can create work disincentives in some cases
- Must balance support needs against incentive effects

We examined several programs through the labour supply model:

- Demogrants, welfare, NIT, wage subsidies, tax credits
- Employment insurance
- Disability and workers compensation
- Child care subsidies